Program overview

12-Sep-2023 13:49

Year **2023/2024**
Organization **Electrical Engineering, Mathematics and Computer Science**
Education **Minors EWI**

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
ET-Mi-190	**ET-Mi-190-23 Electrical Sustainable Energy Systems**						
ET-Mi-190 for EE students 2023							
Compulsory Courses ET-Mi-190 for EE students							
AE3516A	Fundamentals of Wind Energy I	3					
EE3065TU	Reliability of Sustainable Power Systems	3					
EE3400TU	Sustainable Electrical Distribution Networks	3					
ET3034TU	Solar Energy	3					
ET3036TU	Project Design of Sustainable Energy Supply	6					
ET3037TU	Project Integrating Renewable Energy	6					
Keuzevakken ET-Mi-190 for EE students	**Electives ET-Mi-190 for EE students**						
EE3060TU	Agent-based Energy Markets	3					
EE3105TU	Sustainable Energy Technologies	3					
EE3110TU	Energy Efficiency	3					
EE3390TU	Introduction to Cyber Security for Power Grids	3					
ET-Mi-190 for non-EE students 2023							
Verplicht deel ET-Mi-190 for non-EE students	**Compulsory Courses ET-Mi-190 for non-EE students**						
EE3395TU	Introduction to Electrical Power Engineering	6					
EE3400TU	Sustainable Electrical Distribution Networks	3					
ET3034TU	Solar Energy	3					
ET3036TU	Project Design of Sustainable Energy Supply	6					
ET3037TU	Project Integrating Renewable Energy	6					
Keuzevakken ET-Mi-190 for non-EE students	**Electives ET-Mi-190 for non-EE students**						
AE3516A	Fundamentals of Wind Energy I	3					
EE3060TU	Agent-based Energy Markets	3					
EE3065TU	Reliability of Sustainable Power Systems	3					
EE3105TU	Sustainable Energy Technologies	3					
EE3110TU	Energy Efficiency	3					
EE3390TU	Introduction to Cyber Security for Power Grids	3					

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Minors EWI**

ET-Mi-190	
Minor Coordinator	Dr. P.P. Vergara Barrios
In association with the Faculty of	EWI LR
Program Goals	The Minor Electrical Sustainable Energy Systems (ESES) covers fundamental aspects related to electrical power generation from renewable energy sources, the principles of electric power conversion and storage, energy efficiency, reliability, cyber-security, and intelligent grid management. Therefore, the objectives of the ESES Minor are defined as follows: Provide simple physical tools to estimate the potential of renewable energy sources in reference to electric power supply. Analysis of the performance of systems for transmission, distribution, and storage of electrical energy. Design on paper of a renewable system powered by solar and wind energy. Design of smart grid solutions for sustainability problems.
Administration by the Faculty of	EWI
Administration by the Education of	EWI
Minor Title	Electrical Sustainable Energy Systems
EC Minor (Volume)	30
Contact for Students (Minor)	Dr. ir. José Rueda
Intended for	Students who have successfully completed the second year of a BSc programme at TU Delft or equivalent level at other universities.
Expected prior Knowledge	Mathematics (numerical analysis, linear algebra, complex numbers, calculus), elementary physics, basic principles of circuit theory, electronic circuits, and programming (e.g. Matlab/Python).
Minor Coherence / Goal	After completing ESES Minor, the students acquire good insight into the basic principles of the components and the operation of an interconnected electrical sustainable power system. This is relevant for students interested in pursuing a master programme related to this area. Please note that attending and completing ESES Minor does not entail a guarantee an admission in master programmes such Electrical Engineering, Sustainable Energy Technology and the European Wind Energy Master. To this aim, the minor offers a set of theoretical and project based courses that allow the students to learn from the scratch, exercise, and apply the acquired knowledge in near-real-world problems, as well as to develop practical skills in a multidisciplinary context.
Minor content 2	NOTE: The information provided here may change depending on the developments around the coronavirus. The minor offers a number of courses and projects as follows: For non-electrical engineering students: Compulsory courses EE3395TU Introduction to Electrical Power Engineering ET3037TU Project Integrating Renewable Energy EE3400TU Sustainable Electrical Distribution Networks ET3034TU Solar Energy ET3036TU Project Design of sustainable energy supply Elective courses EE3065TU Reliability of Sustainable Power Systems AE3516A Fundamentals of Wind Energy I EE3105TU Sustainable Energy Technologies EE3060TU Agent-based energy markets EE3110TU Energy Efficiency EE3390TU Introduction to Cyber Security for Power Grids For electrical engineering students: Compulsory courses EE3065TU Reliability of Sustainable Power Systems ET3037TU Project Integrating Renewable Energy AE3516A Fundamentals of Wind Energy I EE3400TU Sustainable Electrical Distribution Networks ET3034TU Solar Energy ET3036TU Project Design of Sustainable Energy Supply Elective courses EE3105TU Sustainable Energy Technologies EE3060TU Agent-based energy markets EE3110TU Energy Efficiency EE3390TU Introduction to Cyber Security for Power Grids All lectures of this minor are given in English.
Minor Assessment	All courses and projects of the minor are assessed separately, and the minimum passing grade for each one is 6. The minor is successfully done if all passed courses jointly yield 30 ECTs.
Maximum number of participants	50
Minimum number of participants	30

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Minors EWI**

ET-Mi-190 for EE students 2023	
Minor Coordinator	Dr. P.P. Vergara Barrios

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Minors EWI**

Compulsory Courses ET-Mi-190 for EE students

AE3516A	Fundamentals of Wind Energy I	3
Responsible Instructor	Dr.ir. A.C. Viré	
Instructor	Dr. D.A. von Terzi	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course introduces the topic of offshore wind energy by going through the main steps that a wind farm developer needs to achieve in order to build an offshore wind farm. The main topics cover in the course are: - Historical developments in wind energy - Wind energy harvesting and Betz limit - Offshore wind and wave conditions - Estimation of the energy yield - Design considerations for wind turbine farms - Economic aspects - Planning issues - Emerging concepts such as floating wind turbines	
Study Goals	At the end of this course, the students will be able to: - Explain the basic principles of wind energy conversion. - Identify the steps required when developing offshore wind farms. - Differentiate between the challenges and opportunities of offshore and onshore wind energy harvesting. - Use existing tools for estimating the power produced by an offshore wind farm and the effect of wakes. - Integrate knowledge from various fields of engineering related to the development of offshore wind parks. - Reflect on the current and future trends in the field of offshore wind energy.	
Education Method	The teaching method uses the principle of flipped classroom. This means that, every week, the students have to perform online assignments (e.g. watch a lecture, read a text, etc) prior to coming to class. In class, additional lecture materials, exercises and feedback are given to the students.	
Assessment	The final assessment is a written exam. Additional points are given for performing the online activities prior to class.	

EE3065TU	Reliability of Sustainable Power Systems	3
Responsible Instructor	Dr.ir. J.L. Rueda Torres	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Solid background in calculus and linear algebra. Basics of probability, statistics and Matlab programming.	
Course Contents	This course focusses on applied probability theory for reliability analysis of sustainable electrical power systems. It starts with a review of the basics of probability theory, the definition of reliability and recent developments of deterministic and probabilistic reliability analysis. As accurate failure statistics are the base of any reliability calculation, the practices for collection and modelling of failure statistics will also be addressed. Next, various reliability assessment methods for smaller systems (e.g. reliability networks, fault-tree analysis, Markov models), and for larger networks (e.g. generation adequacy, state enumeration, Monte Carlo simulation) are introduced and discussed. Besides, the application of the studied models and methods on various real systems will be carried out in order to assess risks for security of supply as well as to elucidate possible mitigation measures.	
Study Goals	After attending this course, the students are able to: - Describe the concept of power system reliability and place it in the context of recent developments and future challenges. - Collect and accurately model failure statistics. - Evaluate the reliability of small and large systems. - Define countermeasures to mitigate reliability threats.	
Education Method	Lectures: 7x 2 hours Practicums sessions: 7x 2 hours Examination: 3 hours Self-study: ~53 hours (~7x6 hours/week + 11 hours exam preparation)	
Literature and Study Materials	Book: Bart W. Tuinema, José L. Rueda Torres, Alexandru I. Stefanov, Francisco M. Gonzalez-Longatt, and Mart A. M. M. van der Meijden, "Probabilistic Reliability Analysis of Power Systems," Springer Nature Switzerland AG 2020, May 2020. DOI: https://doi.org/10.1007/978-3-030-43498-4. Print ISBN: 978-3-030-43497-7. Online ISBN: 978-3-030-43498-4. or Reader: Bart W. Tuinema, José L. Rueda Torres, and Mart A.M.M. van der Meijden, Reliability of Sustainable Power Systems Course Reader Delft University of Technology, 2018.	
Assessment	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) The assessment consists of a digital exam/re-exam and reports of the practicum sessions. The students will make short reports of 6 out of 7 practicums (i.e. the Matlab tutorial given in the first session is excluded).	
Exam Hours	There will be a digital examination in the exam period of Q1. Duration: 3 hours. The digital resit is in the exam period of Q2. Duration: 3 hours.	
Permitted Materials during Tests	Book, course notes and formula sheet are not allowed. It is allowed to use a pocket calculator for the calculations. All other electronic devices (e.g. smartphone, tablet, notebook) must be turned off and never be used during the exam.	
Judgement	The final grade is calculated as 0.7x the exam/re-exam grade + 0.3x the average grade of the practicum reports. The maximum achievable grade is 10 points. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken.	
maximum aantal deelnemers	70	
Co-Instructor	Dr. S.H. Tindemans	

EE3400TU	Sustainable Electrical Distribution Networks	3
Responsible Instructor	Dr. P.P. Vergara Barrios	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Solid background in calculus and linear algebra, which should be acquired during the first two years of the BSc program. Basic experience Any programming language. The course provides a guideline to self-learn Python from scratch. Optional prerequisite: Introduction to Electrical Power Engineering, EE3395TU.	
Parts	NOTE: This is a new course. The final lecture structure and/or content may slightly differ from the description presented below. Theory Lecture 1: Introduction, reinforcing background and contextualization. Lecture 2: Symmetrical components and lines modeling Lecture 3: Transformer modeling and distribution networks analysis Lecture 4: Distribution system simulators (DSS). Lecture 5: Evaluation of state performance Lecture 6: Distribution networks with distributed energy resources. Lecture 7: Distribution network planning and expansion. Lecture 8: Invited lecture. Exercises Practicum 1: Exercises related to Lecture 1 Practicum 2: Exercises related to Lecture 2 Practicum 3: Exercises related to Lecture 3 Practicum 4: Exercises in a Distribution System Simulator Practicum 5: Exercises in a Distribution System Simulator Practicum 6: Final Project working session Practicum 7: Final Project working session Practicum 8: Final Project working session	
Course Contents	The term sustainable electrical systems are usually used to characterize the undergoing transformation of transmission and distribution networks with increasing integration of sustainable energy resources. In operational terms, transmission networks' physical properties and associated operational challenges are different from those that can be seen in distribution networks. In this course, students will develop the necessary technical skills to understand, analyze and propose innovative solutions to solve the expected technical problems that distribution networks face. To do this, we will first introduce and contextualize sustainable distribution networks. We will highlight the main modeling and challenges when compared with large transmission systems. Then, we will learn how to model distribution networks using a three-phase power flow formulation. As usually these formulations are too complex to solve by hand, we will use professional distribution system simulators to perform more complex engineering analysis. Finally, we will study the fundamentals/rationale of generic methodologies/approaches for planning, operation, and expansion of distribution networks with large numbers of PV systems, electric vehicles, electric heat pumps.	
Study Goals	At the end of the Electrical Sustainable Distribution Networks course, the students will be able to: 1) Model a sustainable distribution network and its components (e.g., transformers, cables, voltage regulators, capacitor banks) using a multi-phase power flow formulation. 2) Solve a multi-phase power flow formulation using a distribution system simulation (DSS) software. 3) Design solution strategies to overcome technical problems (e.g., overvoltage, transformer overloading, line congestions) present in sustainable distribution networks. 4) Apply and interpret the assessment outcomes obtained by sustainable distribution network planning and expansion techniques to accommodate increasing consumption and distributed energy resources (e.g., PV systems, electric vehicles, electric heat pumps).	
Education Method	Attendance to lectures: 7x2 hours/lecture. Total: 14 hours Attendance to practicums: 6x2 hours/practicum. Total: 16 hours Examination: 3 hours Self-study: 59 hours (7x5 hours/week + 22 hours exam preparation)	
Books	William H. Kersting, Distribution System Modeling and Analysis, Fourth Edition, CRC Press, Taylor & Francis Group, 2018. Francisco Gonzales-Longatt, Jose L. Rueda Torres, Advanced Smart Grid Functionalities Based on PowerFactory, Springer, 2018.	
Assessment	Final exam, quiz based on the practicums and final software-based graded Project. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024	
Exam Hours	The digital exam is in the exam period of Q2. Duration: 2 hours. The digital resit is in the exam period of Q3. Duration: 2 hours.	
Judgement	The digital exam/re-exam is graded on a scale of 0-10 points (i.e., the maximum achievable grade). The course also has three exercise-based and two software-based Practicums. After each Practicum, you must take a mandatory quiz that can have up to 5 exam-type questions. The questions are based on the results that you will obtain after solving the Practicums. All practicums will be graded on a scale of 0-10 points. Practicum 6 introduces the Project, which is a software-based exercise. The Project will be graded on a scale of 0-10 points. A final (short) report must be delivered explaining your developments. The final exam corresponds to 50% of the final grade, the Practicum quizzes 20%, and the Project report 30%.	
Co-Instructor	Dr.ir. J.L. Rueda Torres	

ET3034TU	Solar Energy	3
Responsible Instructor	Prof.dr.ir. A.H.M. Smets	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In the course Solar Energy, you will learn to design a complete photovoltaic (PV) system for any application and location. This course introduces the technology that converts solar energy into electricity. The role of solar energy in both the energy transition towards a sustainable future and climate change mitigation will be discussed in detail. The physical principle of the PV energy conversion together with the design rules for a solar cell are introduced. These design rules are applied on the various PV technologies from cell up to module level. This program highlights various metrics related to performance, costs, reliability and circularity that will be used to evaluate the advantages and limitation of different PV technologies and its applications.	
Study Goals	After completion of the course, the students are ablo to: a) Reflect on the role of solar energy in both the energy transition towards a sustainable future and climate change mitigation b) Understand the physical working principles of photovoltaic energy conversion in solar cells c) Recognize and describe the various photovoltaic technologies and evaluate their applicability using various metrics related to performance, costs, reliability and circularity of PV. d) Design a complete photovoltaic system for any particular application and location	
Education Method	The education method is the so-called flipped-class room. Lectures will be provided in small video lectures on Brightspace. The on-campus lectures are focused on problem-solving. Students will actively work on problems that cover the entire content of the course and the various engineering skills. Solutions will be discussed during the lectures. Homework for every lecture includes: watching the relevant video lectures dedicated to the upcoming course and the remaining exercises of the previous lecture.	
Literature and Study Materials	Literature and Study Materials are provided on Brightspace: Videolectures Lectureslides. Large set of problems & exercises including answers. Answers are posted on Brightspace after the problems have been discussed in the lectures. Large set of examples of exams and answers. The book 'Solar Energy, basics, technology and systems' is free to download as e-book via all online retailers. Hardcopies of the book are sold at the ETV counter (building 36).	
Assessment	The course ET3034TU Solar Energy Energy will be assessed in Q2 and in Q3 (resit) of the academic year 2022/2023 as follows: The exam is a written exam. The exam takes 3 hours. The exam consist of open questions and multiple choice questions. A formula sheet will be provided during the exam (example is poste on Brightspace). You can use only pen, paper and a calculator. The exam will be assessed with a mark with a resolution of 0.5. To pass the exam a score equal to or higher than 58.0 out of 100.0 is needed.	
Remarks	Coordinator: Arno Smets	

ET3036TU	Project Design of Sustainable Energy Supply	6
Responsible Instructor	Dr. S.H. Tindemans	
Co-responsible for assignments	N.K. Panda	
Co-responsible for assignments	N. Li	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Courses in Q1: ET3365TU-D1 Introduction to Electrical Power Engineering -Part 1 EE3105TU Sustainable Energy Technologies	
Course Contents	In this project-based course, students will investigate the value of combining a portfolio of energy resources to satisfy the needs of consumers in a small energy system. The variability of supply and demand, the use of storage and the trade-offs of interconnecting with a larger system are investigated, using purpose-built simulation software.	
Study Goals	After completing this course, the students will be able to: - Describe key steps in computer aided modelling and simulation of the energy balance of sustainable electrical energy systems. - Translate service requirements into design specifications for small energy systems. - Construct a model of a small electrical energy system in purpose-built software, making use of a provided component library. - Simulate the system under time-varying operating conditions and analyse its performance. - Design and test a control strategy for electrical energy storage devices to cope with surplus/deficit of renewable energy. - Choose type and sizing of components to deliver adequate system performance on multiple performance metrics.	
Education Method	This is a group-based project that is supported by 8 lab sessions that are centred on computer aided modelling and simulation.	
Literature and Study Materials	Materials provided on Brightspace: tutorials, project guidance, relevant papers. During this project, students will search for additional literature that is relevant to their project.	
Assessment	Students are allocated to multi-disciplinary groups at the beginning of the course. At the end of the course, each group submits: - Item 1: The (computer) design of the electrical energy system. - Item 2: A report in PDF format. - Item 3: Group presentation (oral defense). Each item is graded on the scale 0-10. The final grade is the weighted average of the model (20%), report (50%) and presentation (30%). Individual marks are rounded to 0.1 and the total is rounded to the nearest unit of 0.5. A minimum mark of 5.0 is required for each component (model, report, presentation) in order to pass the course. Group members must perform equitably throughout the project, report writing and during the oral defence. Final grades are awarded based on the group's performance: all participating group members are awarded the same grade. If there is evidence of highly unequal contributions by group members, individual adjustments can be made to the final grade, in the range of -2 (for severe underperformance) to +1 for an exceptional individual contribution. If some or all group members obtain a mark between 4.0 and 5.7 for one or more components, a repair opportunity will be offered, with which a maximum mark of 6.0 can be obtained for the respective component(s). This will take place in the first half of Q3 (after the course in Q2). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	The group presentation will take place in the exam period of Q2. Duration of the presentation of each group: 30 minutes. A presentation schedule will be announced during the course. Students only attend the presentation of their own group. All members of each group must be present during the group presentation. In case of unjustified absence, a member will receive a 0 for their presentation if they did not take part in preparations; a mark of 4.0 is awarded for absence after full participation in preparations for the presentation.	
Permitted Materials during Tests	Powerpoint presentations and other visual aids are permitted during the group presentation.	
Remarks	- The entire course is given in English. - No theory lectures, except for an introductory briefing. The focus is on practical aspects and computer-aided engineering. - The students will be grouped into several multidisciplinary teams. - The course features challenges and working environment similar to real-world engineering projects. - A 2-hour tutorial on the simulation platform is planned for the first lab session.	
maximum aantal deelnemers	50	

ET3037TU	Project Integrating Renewable Energy	6
Responsible Instructor	Dr.ir. R. Vismara	
Responsible Instructor	Prof.dr.ir. O. Isabella	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic knowledge of solar energy and electrolyzers/fuel cells.	
Course Contents	Sustainable energy experiments focused on (i) PV module characteristics, (ii) PV module configurations and systems, (iii) off-grid and grid-connected PV systems, (iv) PV system electronics, (v) electrolyzers and fuel cells, and (vi) green hydrogen production and utilization	
Study Goals	By the end of the course, you should be able to: (1) Perform laboratory measurements with the appropriate equipment (2) Describe the characteristics of solar cells, electrolyzers, and fuel cells (3) Recognize the influence of the light and temperature on the characteristics of solar cells (4) Identify the optimal operating point of photovoltaic systems (5) Fabricate a simple DC/DC converter (6) Demonstrate the operation of a DC/DC converter and maximum power point tracking (7) Quantify the losses in photovoltaic + hydrogen systems (8) Compose well-written scientific reports.	
Education Method	Laboratory sessions based on measurements and design	
Literature and Study Materials	Study guides and didactical material are available via Brightspace.	
Assessment	The final grade is the average of three reports. A sufficient grade (> 5.8) in all three reports is required to successfully complete the course Max 1 retry session and re-submission are allowed Failure of reports: - 1 report: use buffer session in weeks 1.8-1.9 - 2 or 3 reports: failure of the entire course Three areas are assessed: - Content: 65% of the final grade (literature review, solving tasks, data processing, discussion) - Layout: 25% of the final grade (structure, data visualization: clear graphs, images, tables, etc.) - Language: 10% of the final grade (clarity, grammar mistakes) --- IMPORTANT: A virtual version of this course is available for students to attend in Q2, Q3, and Q4 of the academic year 2022-2023. Each of the six tasks of the on-campus course will run in the browser of your choice, in the form of a web application accessible via Brightspace. The assessment of the online version is entirely automatic, by means of numerical exercises and multiple-choice questions. A sufficient grade (> 57.5/100) in all six tasks is necessary to complete the course. The final grade is the average of the six grades. A maximum of one retry session is allowed. No final written or oral exam is planned for this course. This assessment procedure applies to any future virtual instances of the course (i.e. not for the on-campus instance).	
Remarks	This practical course will be given on campus only in Q1. On the other hand, a virtual version of the course (Virtual Project Integrating Renewable Energy) will be available in Q2, Q3, and Q4. Both the enrolment and the access to the web application are made available via Brightspace from Q1 of the academic year 2022-2023.	
Co-Instructor	Dr.ir. R. Vismara	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Minors EWI**

Keuzevakken ET-Mi-190 for EE students

EE3060TU	Agent-based Energy Markets	3
Responsible Instructor	Prof.dr.ir. J.A. la Poutré	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course focuses on balancing supply and demand in future energy systems, especially addressing smart grids, renewable resources, and heavy energy consuming devices. The focus is especially on the role of ICT systems (Information and Communication Technology) and markets therein. The course starts with an overview of principles of smart energy systems and addresses the uncertainty of energy supply by renewable resources, the increasing intensity of demand by e.g. e-vehicles and heat pumps, the limited capacity of distribution networks, and the principle of supply-driven demand. Next, important techniques from power system economics and ICT for balancing supply and demand are addressed, for proper allocation of energy, especially based on multi-agent systems (based on software agents) and economic market mechanisms. Several solutions and approaches are considered, both fundamental (theoretical) as well as practical (applied), and especially considering agent-based approaches. Also, requirements and constraints for the ICT infrastructure are addressed, in order to enable such ICT solutions. In addition, societal aspects are considered, like security of supply, fairness between users, privacy of users, and roles of stakeholders in the energy system. Approaches and techniques of the current state of the art are discussed and compared. Additional aspects for future innovation of the current technologies (especially ICT) are discussed, especially based on requirements in the energy, societal, and business domains.	
Study Goals	L01: Describe the principles of smart energy systems with respect to uncertainty of energy supply and supply-driven demand L02: Apply basic techniques from the power system economics, multi-agent systems, and ICT domains, especially with respect to automated allocation and market mechanisms for energy systems L03 Describe fundamental as well as practical solutions for supply/demand matching in future energy systems L04: Relate solutions in the context of societal requirements like security of supply, privacy of users, and fairness between users L05: Evaluation of solutions with respect to ICT aspects and energy aspects, as well as business, economics, and societal aspects, and can formulate possible desires and innovation possibilities for future solutions.	
Education Method	Theory lectures (2 online hours/week), self-study	
Assessment	Written exam. NOTE: The information provided here may change depending on future developments around the coronavirus.	

EE3105TU	Sustainable Energy Technologies	3
Responsible Instructor	H. Ziar	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	A transition to sustainable energy is needed for our climate and welfare. This course is an introduction to the Master Programme Sustainable Energy Technology at TU Delft and is aimed at Bachelor students from science and engineering disciplines. In this course, you will learn how to assess the potential for energy-use reduction and the potential of generating and storing renewable energy sources such as wind, solar and hydro. Youll learn how the flow of energy from various sources interact with different consumption sectors and how power plants participate in electricity markets. Course content: 1. Climate change and role of sustainable energies 2. Energy access and consumption of energy 3. Generation of renewable energy: Hydro-power, wind, and solar PV 4. Energy Balances 5. Electricity Markets 6. Storage of energy	
Study Goals	Learning objectives: LO1: Describe the reasoning behind global warming and link to the need for sustainable energy technologies. LO2: Assess resources potential, energy use, and ways to reduce consumption at different sectors (transport, industry, buildings) LO3: Calculate electricity generation from various renewable energy plants, such as wind, solar, and hydro LO4: Analyze energy balance sheets, link them with Sankey diagrams and assess future energy policies LO5: Rank merit order of generation and calculate power plant earnings in electricity market LO6: Calculate the capacity of various energy storage technologies, such as flywheel, pump-hydro, compressed air, battery, and super-capacitors	
Education Method	Lectures/instructions, video lectures, Exercises	
Assessment	The course EE3105TU Sustainable Energy Technologies will be assessed in Q1 and in Q2 (resit) of the academic year as follows: The exam is a written on-campus exam. The exam consists of open questions and multiple-choice questions. The exam is a closed book but you can bring one page of formula sheet. Like the conventional on-campus exam, you can use only pen, paper and a calculator. The exam takes 3 hours. Communication with others is not allowed! The exam will be assessed with a mark with a resolution of 0.5. A score equal to or higher than 58.0 out of 100 is needed to pass the exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Tags	Economics Energy Energy & Industry Industry Physics Sustainability Technology	

EE3110TU	Energy Efficiency	3
Responsible Instructor	Prof.dr. P. Palensky	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	nothing	
Course Contents	Energy efficiency is the primary means to achieve our climate and emission goals. Every kWh wasted is one too much. This course communicates how to achieve efficiency in the energy system by working with dedicated methods. The first step is typically to define goals and to quantify the efficiency targets. This differs from country to country and between sectors and also depends on the stakeholders (governments, city administration, industrial company, etc.). Once goals are clear, it is needed to get data. A system can only be improved if the details are known and understood. Getting the required data involves energy metering, sensor networks, database access, municipal registries, business data, supply chain and manufacturing execution systems and other sophisticated sources of data. The heterogeneity of the data requires pre-processing before it can be used for benchmarking or KPI purposes. After the data is converted into information it needs to be visualized, correlated, and communicated to the stakeholders. This course will cover the basic methods to accomplish this. Stakeholders that are informed and motivated to improve efficiency will then implement measures. The classical way to improve efficiency is to replace old, inefficient equipment (e.g. windows, machines, insulation) with new, better ones. Additional to that, controls have a large influence on the efficiency of individual equipment and entire systems. Implementing improvement measures is not the last step. An entire, and recurrent, process of managing efficiency measures must be implemented. Periodic measurements are needed to verify the success of the measures and to adjust settings that are not ideal.	
Study Goals	LO1: Analyze and improve an energy-relevant technology/process. LO2: Conduct an energy efficiency audit of an industrial or commercial customer. LO3: Calculate and visualize energy analysis data.	
Education Method	Depending on the regulations either frontal lecture on campus or virtual lecture with pre-recorded videos. Theory lectures, quizzes: Fundamentals Assignment 1: efficiency study, report and slides, presentation Assignment 2: energy audit executive report	
Literature and Study Materials	Slides and Videos available on Brightspace	
Books	Some related literature (not needed for the lecture, just for reference!): Clive Beggs; Energy: Management, Supply and Conservation; Routledge Publishers, 2009; ISBN: 978-0750686709 D. Yogi Goswami / Frank Kreith; Handbook of Energy Efficiency and Renewable Energy; CRC Press 2007; ISBN: 978-0849317309, Kornelis Blok / Evert Nieuwlaar; Introduction to Energy Analysis; Tylor and Francis, 2016; ISBN: 978-1-138-67115-7	
Reader	see Brightspace	
Assessment	Assignment Total weighs 60% Exam Total weighs 40% Details see slides	
Permitted Materials during Tests	Paper/pencil, calculator	
Tags	Energy Energy & Industry Integrated Modelling Sustainability	

EE3390TU	Introduction to Cyber Security for Power Grids	3
Responsible Instructor	Dr. A.I. Stefanov	
Instructor	Dr. A.I. Stefanov	
Contact Hours / Week x/x/x/x	0/2/0/0 HC & PT	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic knowledge of circuit theory and power system components is preferred although not necessary.	

Summary

The energy system transition towards a sustainable, low carbon future is led by decarbonisation, decentralisation, and digitalisation. Power grids are undergoing a fast-pace process of digitalisation for enhanced monitoring and control capabilities and grid intelligence. However, the increased integration of digital technologies, such as the next generation of Operational Technologies (OTs) and digital substations, implies a new risk as ITOT systems for smart grids are vulnerable to cyber attacks.

Examples of cyber security incidents related to power grids already exist around the world. On December 23, 2015 cyber attacks were conducted on the power grid in Ukraine that resulted in power outages, which affected 225,000 customers. More sophisticated cyber attacks on the Ukrainian power grid followed on December 17, 2016 resulting in a power outage in the distribution network where 200 MW of load was unsupplied.

The complexity of cyber attacks on power systems is on the rise. They can initiate cascading failures and result in a catastrophic blackout. A power outage can disrupt the entire energy chain including water supply, heating, and gas networks. Without power, hospitals and other critical services are severely affected. A disruption of service may lead to financial loss, damages, chaos, or even a loss of lives.

With respect to security of supply and reliability of the future energy system provision, special attention is needed for new vulnerabilities and cyber threats that come with digitalisation. Accordingly, the cyber security aspects are critical for a further power grid digitalisation.

This course aims to analyse and enhance the cyber security of smart grids. It analyses the typical ITOT system architectures for power grid monitoring and control and explains the main cyber security concepts. The course provides essential knowledge to analyse cyber security threats, advanced persistent threats, threat vectors, and vulnerabilities of IT OT systems for power grids. It examines the kill chain stages of the cyber attack process on industrial control systems using real world study cases. The course studies the state-of-the-art security controls and best cyber security practices to defend against cyber attacks and effectively mitigate their impact on power system operation.

Course Contents

This course focuses on cyber security of Information Technology (IT) Operational Technologies (OT) networks used for power system operation. It starts with an overview of the smart grid and its industrial control systems (EMS/SCADA) and introduces computer networks and cyber security.

Next, the course will introduce the cyber kill chain method for cyber security investigation and intelligence-driven defence. Kill chains are used to understand, anticipate, recognise, and combat cyber attacks and advanced persistent threats. The course uses the kill chains of the 2015/2016 cyber attacks on Ukrainian power grid as real-world study cases throughout the course.

The main IT-OT system vulnerabilities and cyber threats are analysed. The course covers threats such as passive and active reconnaissance, social engineering, and malware. It investigates various threat vectors on host and network security and advanced cyber attacks targeting industrial control systems. The impact on power system operation is analysed.

The course discusses the state-of-the-art security controls to defend and mitigate cyber attacks such as access controls, authentication management, cryptography, firewalls, secure communication protocols, network segmentation, and intrusion detection and prevention systems. The best practices to improve power grid resilience to cyber attacks are discussed.

Study Goals

Learning objectives:
At the end of the course, the students will be able to:
1) Analyse cyber security threats, threat vectors, and vulnerabilities of ITOT systems for power grids.
2) Examine kill chain stages of the cyber attack process and analyse the impact of cyber attacks on power system operation.
3) Recommend appropriate security controls to mitigate cyber attacks and describe best practices to enhance the cyber security of power grids.

Education Method

Lectures: theory and real-world study cases (8 x 2h/lecture)
Practicums: hands-on computer labs (4 x 2h/practicum)

Literature and Study Materials

Book: Chwan-Hwa Wu & J. David Irwin. (2013). Introduction to computer networks and cybersecurity. CRC Press. ISBN: 9781498760133

Assessment

The grade is formed based on the following elements:
30% from assignments (four practicum reports submitted on Brightspace)
70% final digital exam, online exam or online oral exam depending on covid-19 restrictions

All assignments and reports are mandatory.

Disclaimer: information may change depending on the developments around the coronavirus.

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Minors EWI**

ET-Mi-190 for non-EE students 2023	
Minor Coordinator	Dr. P.P. Vergara Barrios

Verplicht deel ET-Mi-190 for non-EE students

EE3395TU	Introduction to Electrical Power Engineering	6
Responsible Instructor	Dr.ir. J.L. Rueda Torres	
Instructor	Dr. Z. Qin	
Instructor	G.R. Chandra Mouli	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Solid background in calculus, linear algebra, and elementary physics.	
Parts	Theory Topic 1: Fundamentals of electric circuits Topic 2: Electric power, transformers and three-phase systems Topic 3: DC and AC electric machines Topic 4: Power electronics and motor drives Topic 5: Interconnected power systems Topic 6: Generation, utilization, and balancing of electric power Topic 7: Power system control and power flow analysis Exercises Practicum 1-7: Exercises related to topic 1-7, respectively	
Course Contents	This course teaches the fundamentals and analyses tools for the study of electrical sustainable power systems. The physics governing electric energy quantities such as voltage, electrical current, impedance, and power are explained. The working principles of the most important electrical energy components such as transformers, electric machines, power electronic devices and converters are given and their mathematical models are derived. Next, the rationale of interconnected electrical sustainable power systems and per-unit normalization for systemic studies are overviewed. The behavior and systemic interplay of power generation, transmission, and consumption are analyzed. The final stage covers the basics of power system control and power flow analysis based on iterative methods.	
Study Goals	After attending this course, the students are able to: 1) Describe passive network elements, phasor representation, and the laws of Kirchhoff, Lorentz, Faraday. 2) Evaluate voltages, currents, and powers in AC and DC circuits. 3) Recognize the working principles of DC systems and AC systems. 4) Analyse electromechanical energy conversion in DC, synchronous and induction machines. 5) Characterize the structure and operation of interconnected sustainable power systems. 6) Assess the steady-state performance of interconnected sustainable power systems. 7) Synthesize solutions for operational problems in interconnected sustainable power systems.	
Education Method	Attendance to lectures: 14x2 hours/lecture. Total: 28 hours Attendance to practicums: 7x2 hours/practicum. Total: 14 hours Exam/re-exam: 3 hours Self-study: 124 hours (7x14 hours/week + 26 hours exam/re-exam preparation)	
Books	Topics 1-4: Giorgio Rizzoni, and James Kearns, Principles and Applications of Electrical Engineering Sixth Edition, McGraw-Hill, April 2015. ISBN: 9780071254441. Topics 5-7: Pieter Schavemaker, and Lou van der Sluis, Electrical Power System Essentials - Second Edition", John Wiley and Sons, Aug. 2017. ISBN: 978-1-118-80347-9.	
Prerequisites	Solid background in calculus, linear algebra, and elementary physics.	
Assessment	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Written exam. Weekly (optional) online tests.	
Exam Hours	The written exam is in the exam period of Q1. Duration: 3 hours. The written re-exam is in the exam period of Q2. Duration: 3 hours.	
Permitted Materials during Tests	Book and course notes are not allowed. Formula sheet (both sides of an A4 paper) is allowed. It is allowed to use a pocket calculator or a graphing calculator for the calculations. All other electronic devices (smartphone, tablet, notebook, USB sticks) must never be used during the exam.	
Maximum number of participants	70	
Judgement	The written exam/re-exam entails 10 points (i.e. the maximum achievable grade). The students will have an optional online test after each practicum. Each test consists of solving exercises and answering theory questions. For each test, a maximum bonus of 0.2 bonus points can be obtained. The collected bonus (maximum 1.4 extra points in total if all assignments are done) will be added to the exam/re-exam grade. It is important to consider that the maximum achievable grade of 10 cannot be exceeded. In case of obtaining a total grade (exam/re-exam + bonus) lower than 6/10, all parts of the course (exam/re-exam and optional online tests) need to be retaken.	
maximum aantal deelnemers	30	

EE3400TU	Sustainable Electrical Distribution Networks	3
Responsible Instructor	Dr. P.P. Vergara Barrios	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Solid background in calculus and linear algebra, which should be acquired during the first two years of the BSc program. Basic experience Any programming language. The course provides a guideline to self-learn Python from scratch. Optional prerequisite: Introduction to Electrical Power Engineering, EE3395TU.	
Parts	NOTE: This is a new course. The final lecture structure and/or content may slightly differ from the description presented below. Theory Lecture 1: Introduction, reinforcing background and contextualization. Lecture 2: Symmetrical components and lines modeling Lecture 3: Transformer modeling and distribution networks analysis Lecture 4: Distribution system simulators (DSS). Lecture 5: Evaluation of state performance Lecture 6: Distribution networks with distributed energy resources. Lecture 7: Distribution network planning and expansion. Lecture 8: Invited lecture. Exercises Practicum 1: Exercises related to Lecture 1 Practicum 2: Exercises related to Lecture 2 Practicum 3: Exercises related to Lecture 3 Practicum 4: Exercises in a Distribution System Simulator Practicum 5: Exercises in a Distribution System Simulator Practicum 6: Final Project working session Practicum 7: Final Project working session Practicum 8: Final Project working session	
Course Contents	The term sustainable electrical systems are usually used to characterize the undergoing transformation of transmission and distribution networks with increasing integration of sustainable energy resources. In operational terms, transmission networks' physical properties and associated operational challenges are different from those that can be seen in distribution networks. In this course, students will develop the necessary technical skills to understand, analyze and propose innovative solutions to solve the expected technical problems that distribution networks face. To do this, we will first introduce and contextualize sustainable distribution networks. We will highlight the main modeling and challenges when compared with large transmission systems. Then, we will learn how to model distribution networks using a three-phase power flow formulation. As usually these formulations are too complex to solve by hand, we will use professional distribution system simulators to perform more complex engineering analysis. Finally, we will study the fundamentals/rationale of generic methodologies/approaches for planning, operation, and expansion of distribution networks with large numbers of PV systems, electric vehicles, electric heat pumps.	
Study Goals	At the end of the Electrical Sustainable Distribution Networks course, the students will be able to: 1) Model a sustainable distribution network and its components (e.g., transformers, cables, voltage regulators, capacitor banks) using a multi-phase power flow formulation. 2) Solve a multi-phase power flow formulation using a distribution system simulation (DSS) software. 3) Design solution strategies to overcome technical problems (e.g., overvoltage, transformer overloading, line congestions) present in sustainable distribution networks. 4) Apply and interpret the assessment outcomes obtained by sustainable distribution network planning and expansion techniques to accommodate increasing consumption and distributed energy resources (e.g., PV systems, electric vehicles, electric heat pumps).	
Education Method	Attendance to lectures: 7x2 hours/lecture. Total: 14 hours Attendance to practicums: 6x2 hours/practicum. Total: 16 hours Examination: 3 hours Self-study: 59 hours (7x5 hours/week + 22 hours exam preparation)	
Books	William H. Kersting, Distribution System Modeling and Analysis, Fourth Edition, CRC Press, Taylor & Francis Group, 2018. Francisco Gonzales-Longatt, Jose L. Rueda Torres, Advanced Smart Grid Functionalities Based on PowerFactory, Springer, 2018.	
Assessment	Final exam, quiz based on the practicums and final software-based graded Project. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024	
Exam Hours	The digital exam is in the exam period of Q2. Duration: 2 hours. The digital resit is in the exam period of Q3. Duration: 2 hours.	
Judgement	The digital exam/re-exam is graded on a scale of 0-10 points (i.e., the maximum achievable grade). The course also has three exercise-based and two software-based Practicums. After each Practicum, you must take a mandatory quiz that can have up to 5 exam-type questions. The questions are based on the results that you will obtain after solving the Practicums. All practicums will be graded on a scale of 0-10 points. Practicum 6 introduces the Project, which is a software-based exercise. The Project will be graded on a scale of 0-10 points. A final (short) report must be delivered explaining your developments. The final exam corresponds to 50% of the final grade, the Practicum quizzes 20%, and the Project report 30%.	
Co-Instructor	Dr.ir. J.L. Rueda Torres	

ET3034TU	Solar Energy	3
Responsible Instructor	Prof.dr.ir. A.H.M. Smets	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In the course Solar Energy, you will learn to design a complete photovoltaic (PV) system for any application and location. This course introduces the technology that converts solar energy into electricity. The role of solar energy in both the energy transition towards a sustainable future and climate change mitigation will be discussed in detail. The physical principle of the PV energy conversion together with the design rules for a solar cell are introduced. These design rules are applied on the various PV technologies from cell up to module level. This program highlights various metrics related to performance, costs, reliability and circularity that will be used to evaluate the advantages and limitation of different PV technologies and its applications.	
Study Goals	After completion of the course, the students are ablo to: a) Reflect on the role of solar energy in both the energy transition towards a sustainable future and climate change mitigation b) Understand the physical working principles of photovoltaic energy conversion in solar cells c) Recognize and describe the various photovoltaic technologies and evaluate their applicability using various metrics related to performance, costs, reliability and circularity of PV. d) Design a complete photovoltaic system for any particular application and location	
Education Method	The education method is the so-called flipped-class room. Lectures will be provided in small video lectures on Brightspace. The on-campus lectures are focused on problem-solving. Students will actively work on problems that cover the entire content of the course and the various engineering skills. Solutions will be discussed during the lectures. Homework for every lecture includes: watching the relevant video lectures dedicated to the upcoming course and the remaining exercises of the previous lecture.	
Literature and Study Materials	Literature and Study Materials are provided on Brightspace: Videolectures Lectureslides. Large set of problems & exercises including answers. Answers are posted on Brightspace after the problems have been discussed in the lectures. Large set of examples of exams and answers. The book 'Solar Energy, basics, technology and systems' is free to download as e-book via all online retailers. Hardcopies of the book are sold at the ETV counter (building 36).	
Assessment	The course ET3034TU Solar Energy Energy will be assessed in Q2 and in Q3 (resit) of the academic year 2022/2023 as follows: The exam is a written exam. The exam takes 3 hours. The exam consist of open questions and multiple choice questions. A formula sheet will be provided during the exam (example is poste on Brightspace). You can use only pen, paper and a calculator. The exam will be assessed with a mark with a resolution of 0.5. To pass the exam a score equal to or higher than 58.0 out of 100.0 is needed.	
Remarks	Coordinator: Arno Smets	

ET3036TU	Project Design of Sustainable Energy Supply	6
Responsible Instructor	Dr. S.H. Tindemans	
Co-responsible for assignments	N.K. Panda	
Co-responsible for assignments	N. Li	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Courses in Q1: ET3365TU-D1 Introduction to Electrical Power Engineering -Part 1 EE3105TU Sustainable Energy Technologies	
Course Contents	In this project-based course, students will investigate the value of combining a portfolio of energy resources to satisfy the needs of consumers in a small energy system. The variability of supply and demand, the use of storage and the trade-offs of interconnecting with a larger system are investigated, using purpose-built simulation software.	
Study Goals	After completing this course, the students will be able to: - Describe key steps in computer aided modelling and simulation of the energy balance of sustainable electrical energy systems. - Translate service requirements into design specifications for small energy systems. - Construct a model of a small electrical energy system in purpose-built software, making use of a provided component library. - Simulate the system under time-varying operating conditions and analyse its performance. - Design and test a control strategy for electrical energy storage devices to cope with surplus/deficit of renewable energy. - Choose type and sizing of components to deliver adequate system performance on multiple performance metrics.	
Education Method	This is a group-based project that is supported by 8 lab sessions that are centred on computer aided modelling and simulation.	
Literature and Study Materials	Materials provided on Brightspace: tutorials, project guidance, relevant papers. During this project, students will search for additional literature that is relevant to their project.	
Assessment	Students are allocated to multi-disciplinary groups at the beginning of the course. At the end of the course, each group submits: - Item 1: The (computer) design of the electrical energy system. - Item 2: A report in PDF format. - Item 3: Group presentation (oral defense). Each item is graded on the scale 0-10. The final grade is the weighted average of the model (20%), report (50%) and presentation (30%). Individual marks are rounded to 0.1 and the total is rounded to the nearest unit of 0.5. A minimum mark of 5.0 is required for each component (model, report, presentation) in order to pass the course. Group members must perform equitably throughout the project, report writing and during the oral defence. Final grades are awarded based on the group's performance: all participating group members are awarded the same grade. If there is evidence of highly unequal contributions by group members, individual adjustments can be made to the final grade, in the range of -2 (for severe underperformance) to +1 for an exceptional individual contribution. If some or all group members obtain a mark between 4.0 and 5.7 for one or more components, a repair opportunity will be offered, with which a maximum mark of 6.0 can be obtained for the respective component(s). This will take place in the first half of Q3 (after the course in Q2). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	The group presentation will take place in the exam period of Q2. Duration of the presentation of each group: 30 minutes. A presentation schedule will be announced during the course. Students only attend the presentation of their own group. All members of each group must be present during the group presentation. In case of unjustified absence, a member will receive a 0 for their presentation if they did not take part in preparations; a mark of 4.0 is awarded for absence after full participation in preparations for the presentation.	
Permitted Materials during Tests	Powerpoint presentations and other visual aids are permitted during the group presentation.	
Remarks	- The entire course is given in English. - No theory lectures, except for an introductory briefing. The focus is on practical aspects and computer-aided engineering. - The students will be grouped into several multidisciplinary teams. - The course features challenges and working environment similar to real-world engineering projects. - A 2-hour tutorial on the simulation platform is planned for the first lab session.	
maximum aantal deelnemers	50	

ET3037TU	Project Integrating Renewable Energy	6
Responsible Instructor	Dr.ir. R. Vismara	
Responsible Instructor	Prof.dr.ir. O. Isabella	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic knowledge of solar energy and electrolyzers/fuel cells.	
Course Contents	Sustainable energy experiments focused on (i) PV module characteristics, (ii) PV module configurations and systems, (iii) off-grid and grid-connected PV systems, (iv) PV system electronics, (v) electrolyzers and fuel cells, and (vi) green hydrogen production and utilization	
Study Goals	By the end of the course, you should be able to: (1) Perform laboratory measurements with the appropriate equipment (2) Describe the characteristics of solar cells, electrolyzers, and fuel cells (3) Recognize the influence of the light and temperature on the characteristics of solar cells (4) Identify the optimal operating point of photovoltaic systems (5) Fabricate a simple DC/DC converter (6) Demonstrate the operation of a DC/DC converter and maximum power point tracking (7) Quantify the losses in photovoltaic + hydrogen systems (8) Compose well-written scientific reports.	
Education Method	Laboratory sessions based on measurements and design	
Literature and Study Materials	Study guides and didactical material are available via Brightspace.	
Assessment	The final grade is the average of three reports. A sufficient grade (> 5.8) in all three reports is required to successfully complete the course Max 1 retry session and re-submission are allowed Failure of reports: - 1 report: use buffer session in weeks 1.8-1.9 - 2 or 3 reports: failure of the entire course Three areas are assessed: - Content: 65% of the final grade (literature review, solving tasks, data processing, discussion) - Layout: 25% of the final grade (structure, data visualization: clear graphs, images, tables, etc.) - Language: 10% of the final grade (clarity, grammar mistakes) --- IMPORTANT: A virtual version of this course is available for students to attend in Q2, Q3, and Q4 of the academic year 2022-2023. Each of the six tasks of the on-campus course will run in the browser of your choice, in the form of a web application accessible via Brightspace. The assessment of the online version is entirely automatic, by means of numerical exercises and multiple-choice questions. A sufficient grade (> 57.5/100) in all six tasks is necessary to complete the course. The final grade is the average of the six grades. A maximum of one retry session is allowed. No final written or oral exam is planned for this course. This assessment procedure applies to any future virtual instances of the course (i.e. not for the on-campus instance).	
Remarks	This practical course will be given on campus only in Q1. On the other hand, a virtual version of the course (Virtual Project Integrating Renewable Energy) will be available in Q2, Q3, and Q4. Both the enrolment and the access to the web application are made available via Brightspace from Q1 of the academic year 2022-2023.	
Co-Instructor	Dr.ir. R. Vismara	

Keuzevakken ET-Mi-190 for non-EE students

AE3516A	Fundamentals of Wind Energy I	3
Responsible Instructor	Dr.ir. A.C. Viré	
Instructor	Dr. D.A. von Terzi	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course introduces the topic of offshore wind energy by going through the main steps that a wind farm developer needs to achieve in order to build an offshore wind farm. The main topics cover in the course are: - Historical developments in wind energy - Wind energy harvesting and Betz limit - Offshore wind and wave conditions - Estimation of the energy yield - Design considerations for wind turbine farms - Economic aspects - Planning issues - Emerging concepts such as floating wind turbines	
Study Goals	At the end of this course, the students will be able to: - Explain the basic principles of wind energy conversion. - Identify the steps required when developing offshore wind farms. - Differentiate between the challenges and opportunities of offshore and onshore wind energy harvesting. - Use existing tools for estimating the power produced by an offshore wind farm and the effect of wakes. - Integrate knowledge from various fields of engineering related to the development of offshore wind parks. - Reflect on the current and future trends in the field of offshore wind energy.	
Education Method	The teaching method uses the principle of flipped classroom. This means that, every week, the students have to perform online assignments (e.g. watch a lecture, read a text, etc) prior to coming to class. In class, additional lecture materials, exercises and feedback are given to the students.	
Assessment	The final assessment is a written exam. Additional points are given for performing the online activities prior to class.	

EE3060TU	Agent-based Energy Markets	3
Responsible Instructor	Prof.dr.ir. J.A. la Poutré	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course focuses on balancing supply and demand in future energy systems, especially addressing smart grids, renewable resources, and heavy energy consuming devices. The focus is especially on the role of ICT systems (Information and Communication Technology) and markets therein. The course starts with an overview of principles of smart energy systems and addresses the uncertainty of energy supply by renewable resources, the increasing intensity of demand by e.g. e-vehicles and heat pumps, the limited capacity of distribution networks, and the principle of supply-driven demand. Next, important techniques from power system economics and ICT for balancing supply and demand are addressed, for proper allocation of energy, especially based on multi-agent systems (based on software agents) and economic market mechanisms. Several solutions and approaches are considered, both fundamental (theoretical) as well as practical (applied), and especially considering agent-based approaches. Also, requirements and constraints for the ICT infrastructure are addressed, in order to enable such ICT solutions. In addition, societal aspects are considered, like security of supply, fairness between users, privacy of users, and roles of stakeholders in the energy system. Approaches and techniques of the current state of the art are discussed and compared. Additional aspects for future innovation of the current technologies (especially ICT) are discussed, especially based on requirements in the energy, societal, and business domains.	
Study Goals	L01: Describe the principles of smart energy systems with respect to uncertainty of energy supply and supply-driven demand L02: Apply basic techniques from the power system economics, multi-agent systems, and ICT domains, especially with respect to automated allocation and market mechanisms for energy systems L03 Describe fundamental as well as practical solutions for supply/demand matching in future energy systems L04: Relate solutions in the context of societal requirements like security of supply, privacy of users, and fairness between users L05: Evaluation of solutions with respect to ICT aspects and energy aspects, as well as business, economics, and societal aspects, and can formulate possible desires and innovation possibilities for future solutions.	
Education Method	Theory lectures (2 online hours/week), self-study	
Assessment	Written exam. NOTE: The information provided here may change depending on future developments around the coronavirus.	

EE3065TU	Reliability of Sustainable Power Systems	3
Responsible Instructor	Dr.ir. J.L. Rueda Torres	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Solid background in calculus and linear algebra. Basics of probability, statistics and Matlab programming.	
Course Contents	This course focusses on applied probability theory for reliability analysis of sustainable electrical power systems. It starts with a review of the basics of probability theory, the definition of reliability and recent developments of deterministic and probabilistic reliability analysis. As accurate failure statistics are the base of any reliability calculation, the practices for collection and modelling of failure statistics will also be addressed. Next, various reliability assessment methods for smaller systems (e.g. reliability networks, fault-tree analysis, Markov models), and for larger networks (e.g. generation adequacy, state enumeration, Monte Carlo simulation) are introduced and discussed. Besides, the application of the studied models and methods on various real systems will be carried out in order to assess risks for security of supply as well as to elucidate possible mitigation measures.	
Study Goals	After attending this course, the students are able to: - Describe the concept of power system reliability and place it in the context of recent developments and future challenges. - Collect and accurately model failure statistics. - Evaluate the reliability of small and large systems. - Define countermeasures to mitigate reliability threats.	
Education Method	Lectures: 7x 2 hours Practicums sessions: 7x 2 hours Examination: 3 hours Self-study: ~53 hours (~7x6 hours/week + 11 hours exam preparation)	
Literature and Study Materials	Book: Bart W. Tuinema, José L. Rueda Torres, Alexandru I. Stefanov, Francisco M. Gonzalez-Longatt, and Mart A. M. M. van der Meijden, "Probabilistic Reliability Analysis of Power Systems," Springer Nature Switzerland AG 2020, May 2020. DOI: https://doi.org/10.1007/978-3-030-43498-4. Print ISBN: 978-3-030-43497-7. Online ISBN: 978-3-030-43498-4. or Reader: Bart W. Tuinema, José L. Rueda Torres, and Mart A.M.M. van der Meijden, Reliability of Sustainable Power Systems Course Reader Delft University of Technology, 2018.	
Assessment	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) The assessment consists of a digital exam/re-exam and reports of the practicum sessions. The students will make short reports of 6 out of 7 practicums (i.e. the Matlab tutorial given in the first session is excluded).	
Exam Hours	There will be a digital examination in the exam period of Q1. Duration: 3 hours. The digital resit is in the exam period of Q2. Duration: 3 hours.	
Permitted Materials during Tests	Book, course notes and formula sheet are not allowed. It is allowed to use a pocket calculator for the calculations. All other electronic devices (e.g. smartphone, tablet, notebook) must be turned off and never be used during the exam.	
Judgement	The final grade is calculated as 0.7x the exam/re-exam grade + 0.3x the average grade of the practicum reports. The maximum achievable grade is 10 points. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken.	
maximum aantal deelnemers	70	
Co-Instructor	Dr. S.H. Tindemans	

EE3105TU	Sustainable Energy Technologies	3
Responsible Instructor	H. Ziar	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	A transition to sustainable energy is needed for our climate and welfare. This course is an introduction to the Master Programme Sustainable Energy Technology at TU Delft and is aimed at Bachelor students from science and engineering disciplines. In this course, you will learn how to assess the potential for energy-use reduction and the potential of generating and storing renewable energy sources such as wind, solar and hydro. Youll learn how the flow of energy from various sources interact with different consumption sectors and how power plants participate in electricity markets. Course content: 1. Climate change and role of sustainable energies 2. Energy access and consumption of energy 3. Generation of renewable energy: Hydro-power, wind, and solar PV 4. Energy Balances 5. Electricity Markets 6. Storage of energy	
Study Goals	Learning objectives: LO1: Describe the reasoning behind global warming and link to the need for sustainable energy technologies. LO2: Assess resources potential, energy use, and ways to reduce consumption at different sectors (transport, industry, buildings) LO3: Calculate electricity generation from various renewable energy plants, such as wind, solar, and hydro LO4: Analyze energy balance sheets, link them with Sankey diagrams and assess future energy policies LO5: Rank merit order of generation and calculate power plant earnings in electricity market LO6: Calculate the capacity of various energy storage technologies, such as flywheel, pump-hydro, compressed air, battery, and super-capacitors	
Education Method	Lectures/instructions, video lectures, Exercises	
Assessment	The course EE3105TU Sustainable Energy Technologies will be assessed in Q1 and in Q2 (resit) of the academic year as follows: The exam is a written on-campus exam. The exam consists of open questions and multiple-choice questions. The exam is a closed book but you can bring one page of formula sheet. Like the conventional on-campus exam, you can use only pen, paper and a calculator. The exam takes 3 hours. Communication with others is not allowed! The exam will be assessed with a mark with a resolution of 0.5. A score equal to or higher than 58.0 out of 100 is needed to pass the exam. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Tags	Economics Energy Energy & Industry Industry Physics Sustainability Technology	

EE3110TU	Energy Efficiency	3
Responsible Instructor	Prof.dr. P. Palensky	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	nothing	
Course Contents	Energy efficiency is the primary means to achieve our climate and emission goals. Every kWh wasted is one too much. This course communicates how to achieve efficiency in the energy system by working with dedicated methods. The first step is typically to define goals and to quantify the efficiency targets. This differs from country to country and between sectors and also depends on the stakeholders (governments, city administration, industrial company, etc.). Once goals are clear, it is needed to get data. A system can only be improved if the details are known and understood. Getting the required data involves energy metering, sensor networks, database access, municipal registries, business data, supply chain and manufacturing execution systems and other sophisticated sources of data. The heterogeneity of the data requires pre-processing before it can be used for benchmarking or KPI purposes. After the data is converted into information it needs to be visualized, correlated, and communicated to the stakeholders. This course will cover the basic methods to accomplish this. Stakeholders that are informed and motivated to improve efficiency will then implement measures. The classical way to improve efficiency is to replace old, inefficient equipment (e.g. windows, machines, insulation) with new, better ones. Additional to that, controls have a large influence on the efficiency of individual equipment and entire systems. Implementing improvement measures is not the last step. An entire, and recurrent, process of managing efficiency measures must be implemented. Periodic measurements are needed to verify the success of the measures and to adjust settings that are not ideal.	
Study Goals	LO1: Analyze and improve an energy-relevant technology/process. LO2: Conduct an energy efficiency audit of an industrial or commercial customer. LO3: Calculate and visualize energy analysis data.	
Education Method	Depending on the regulations either frontal lecture on campus or virtual lecture with pre-recorded videos. Theory lectures, quizzes: Fundamentals Assignment 1: efficiency study, report and slides, presentation Assignment 2: energy audit executive report	
Literature and Study Materials	Slides and Videos available on Brightspace	
Books	Some related literature (not needed for the lecture, just for reference!): Clive Beggs; Energy: Management, Supply and Conservation; Routledge Publishers, 2009; ISBN: 978-0750686709 D. Yogi Goswami / Frank Kreith; Handbook of Energy Efficiency and Renewable Energy; CRC Press 2007; ISBN: 978-0849317309, Kornelis Blok / Evert Nieuwlaar; Introduction to Energy Analysis; Tylor and Francis, 2016; ISBN: 978-1-138-67115-7	
Reader	see Brightspace	
Assessment	Assignment Total weighs 60% Exam Total weighs 40% Details see slides	
Permitted Materials during Tests	Paper/pencil, calculator	
Tags	Energy Energy & Industry Integrated Modelling Sustainability	

EE3390TU	Introduction to Cyber Security for Power Grids	3

Responsible Instructor Dr. A.I. Stefanov

Instructor Dr. A.I. Stefanov

Contact Hours / Week x/x/x/x 0/2/0/0 HC & PT

Education Period 2

Start Education 2

Exam Period 2
3

Course Language English

Expected prior knowledge Basic knowledge of circuit theory and power system components is preferred although not necessary.

Summary

The energy system transition towards a sustainable, low carbon future is led by decarbonisation, decentralisation, and digitalisation. Power grids are undergoing a fast-pace process of digitalisation for enhanced monitoring and control capabilities and grid intelligence. However, the increased integration of digital technologies, such as the next generation of Operational Technologies (OTs) and digital substations, implies a new risk as ITOT systems for smart grids are vulnerable to cyber attacks.

Examples of cyber security incidents related to power grids already exist around the world. On December 23, 2015 cyber attacks were conducted on the power grid in Ukraine that resulted in power outages, which affected 225,000 customers. More sophisticated cyber attacks on the Ukrainian power grid followed on December 17, 2016 resulting in a power outage in the distribution network where 200 MW of load was unsupplied.

The complexity of cyber attacks on power systems is on the rise. They can initiate cascading failures and result in a catastrophic blackout. A power outage can disrupt the entire energy chain including water supply, heating, and gas networks. Without power, hospitals and other critical services are severely affected. A disruption of service may lead to financial loss, damages, chaos, or even a loss of lives.

With respect to security of supply and reliability of the future energy system provision, special attention is needed for new vulnerabilities and cyber threats that come with digitalisation. Accordingly, the cyber security aspects are critical for a further power grid digitalisation.

This course aims to analyse and enhance the cyber security of smart grids. It analyses the typical ITOT system architectures for power grid monitoring and control and explains the main cyber security concepts. The course provides essential knowledge to analyse cyber security threats, advanced persistent threats, threat vectors, and vulnerabilities of IT OT systems for power grids. It examines the kill chain stages of the cyber attack process on industrial control systems using real world study cases. The course studies the state-of-the-art security controls and best cyber security practices to defend against cyber attacks and effectively mitigate their impact on power system operation.

Course Contents

This course focuses on cyber security of Information Technology (IT) Operational Technologies (OT) networks used for power system operation. It starts with an overview of the smart grid and its industrial control systems (EMS/SCADA) and introduces computer networks and cyber security.

Next, the course will introduce the cyber kill chain method for cyber security investigation and intelligence-driven defence. Kill chains are used to understand, anticipate, recognise, and combat cyber attacks and advanced persistent threats. The course uses the kill chains of the 2015/2016 cyber attacks on Ukrainian power grid as real-world study cases throughout the course.

The main IT-OT system vulnerabilities and cyber threats are analysed. The course covers threats such as passive and active reconnaissance, social engineering, and malware. It investigates various threat vectors on host and network security and advanced cyber attacks targeting industrial control systems. The impact on power system operation is analysed.

The course discusses the state-of-the-art security controls to defend and mitigate cyber attacks such as access controls, authentication management, cryptography, firewalls, secure communication protocols, network segmentation, and intrusion detection and prevention systems. The best practices to improve power grid resilience to cyber attacks are discussed.

Study Goals

Learning objectives:
At the end of the course, the students will be able to:
1) Analyse cyber security threats, threat vectors, and vulnerabilities of ITOT systems for power grids.
2) Examine kill chain stages of the cyber attack process and analyse the impact of cyber attacks on power system operation.
3) Recommend appropriate security controls to mitigate cyber attacks and describe best practices to enhance the cyber security of power grids.

Education Method

Lectures: theory and real-world study cases (8 x 2h/lecture)
Practicums: hands-on computer labs (4 x 2h/practicum)

Literature and Study Materials

Book: Chwan-Hwa Wu & J. David Irwin. (2013). Introduction to computer networks and cybersecurity. CRC Press. ISBN: 9781498760133

Assessment

The grade is formed based on the following elements:
30% from assignments (four practicum reports submitted on Brightspace)
70% final digital exam, online exam or online oral exam depending on covid-19 restrictions

All assignments and reports are mandatory.

Disclaimer: information may change depending on the developments around the coronavirus.

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